

Notes & Steps



Key idea

A limit asks: what value does a function approach as x gets closer and closer to a number (or infinity)? If the function has a hole at $x = a$, the limit tells you the y -value the graph was heading toward.

Steps

1. Try direct substitution first. If you get a number, that is the limit.
2. If substitution gives $\frac{0}{0}$ (indeterminate), factorise the numerator and denominator, cancel the common factor, then substitute again.
3. For limits at infinity, identify the highest power of x in the numerator and denominator. Divide every term by that power. As $x \rightarrow \infty$, terms like $\frac{1}{x}$, $\frac{1}{x^2}$, etc. become 0.
4. For the form $\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$, expand and simplify first, then let $h \rightarrow 0$.

Common mistake

Trying to substitute $x = a$ directly when the denominator becomes zero. You must factorise and cancel first. For example, $\frac{x^2 - 4}{x - 2}$ at $x = 2$ is $\frac{0}{0}$, but factorising gives $\frac{(x-2)(x+2)}{x-2} = x+2$, so the limit is 4.

Quick checks

- ▶ $\lim_{x \rightarrow 2} (x + 3) = 5$ (direct substitution)
- ▶ $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3} = 6$ (factorise first)
- ▶ $\lim_{x \rightarrow \infty} \frac{1}{x} = 0$ (millionaire theory: 1 divided by a huge number is tiny)
- ▶ $\lim_{x \rightarrow \infty} \frac{2x + 1}{x + 3} = 2$ (ratio of leading coefficients)

Notes & Steps



Example 1: Factorisation

$$\lim_{x \rightarrow 5} \frac{x^2 - 25}{x - 5}$$

Direct substitution: $\frac{0}{0}$, so factorise.

$x^2 - 25 = (x - 5)(x + 5)$, cancel $(x - 5)$ giving $x + 5$.

Substitute $x = 5$: $5 + 5 = 10$.

Example 2: Limits at infinity

$$\lim_{x \rightarrow \infty} \frac{3x^2 + 2x}{5x^2 - x + 1}$$

Highest power is x^2 . Divide each term by x^2 :

$$\frac{3 + \frac{2}{x}}{5 - \frac{1}{x} + \frac{1}{x^2}}$$

2

1

1

Example 3: The $h \rightarrow 0$ form

$$\lim_{h \rightarrow 0} \frac{(x + h)^2 - x^2}{h}$$

Expand: $(x + h)^2 = x^2 + 2xh + h^2$

$$(x + h)^2 - x^2 = 2xh + h^2 = h(2x + h)$$

Divide by h : $2x + h$

As $h \rightarrow 0$: limit = $2x$.

Try these

1. $\lim_{x \rightarrow 4} \frac{x^2 - 16}{x - 4}$

2. $\lim_{x \rightarrow \infty} \frac{4x^2 + 3x}{2x^2 - 5}$

3. $\lim_{h \rightarrow 0} \frac{2(x + h)^2 - 2x^2}{h}$